بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

Nama : Ailsa Rahma

NIM : 241331002

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Prodi / kelas : D3 Teknik Telekomunikasi / 1A TTE

Nama Dosen : Pak Sardjito

TTD : Ailsa

1). Diketahui : medan listrik arah vertikal = 1,4 MN/C = $1{,}4 \times 10^{6}$ N/C

tinta bermassa = $1{,}3 \times 10^{-10}$ kg bermuatan $1{,}5 \times 10^{-13}$ C

kecepatan = 18 m/s

panjang keping = 1,6 cm = 0,016 m

ditanya : tentukan simpangan vertikal tetes tinta keluar dari sistem keping !

Jawaban

- gaya listrik dan percepatan vertikal

$F = q.E$

$= (1{,}5 \times 10^{-13})(1{,}4 \times 10^{6}) = 2{,}1 \times 10^{-7}$ N

percepatan vertikal ay akibat gaya listrik :

$$a_y = \frac{F}{m} = \frac{2{,}1 \times 10^{-7}}{1{,}3 \times 10^{-10}} = 1{,}615 \times 10^{3} \text{ m/s}^2$$

- $t = \frac{L}{V_x}$

$$= \frac{0{,}016}{18} \approx 8{,}89 \times 10^{-4} \text{ s}$$

- $y = \frac{1}{2} a_y t^2$

$= \frac{1}{2} (1{,}615 \times 10^{3})(8{,}89 \times 10^{-4})^2$

$= \frac{1}{2} (1{,}615 \times 10^{3})(7{,}9 \times 10^{-7})$

$= 0{,}638 \times 10^{-3}$ m $\approx 0{,}64$ mm

2). Diketahui = pada saat mencapai keping positif kecepatan = 5×10^{4} m/s

jarak antar keping = 0,5 cm = 0,005 m

massa elektron = 9×10^{-31} kg

muatan elektron = $-1{,}6 \times 10^{-19}$ C

Ditanya : berapakah beda potensial antara keping positif dan negatif ?

Jawab

- $V = \frac{\frac{1}{2} m v^2}{|q|}$

$$= \frac{\frac{1}{2} (9 \times 10^{-31})(5 \times 10^{4})^2}{1{,}6 \times 10^{-19}}$$

VISION

- $V^2 = (5 \times 10^4)^2$

  $= 25 \times 10^8 = 2{,}5 \times 10^9$

- $\frac{1}{2} mV^2$

  $\frac{1}{2} (9 \times 10^{-31})(2{,}5 \times 10^9) = 11{,}25 \times 10^{-22}$

- $V = \frac{11{,}25 \times 10^{-22}}{1{,}6 \times 10^{-19}} \approx 7{,}03\ V$

3). Diketahui = kawat = 2,0 mm = 0,002 m

arus listrik = 800 mA = 0,8 A

Ditanya : Berapa kuat medan listrik ?

Jawaban

$A = \pi r^2$

$= \pi (0{,}002)^2 = \pi \times 4 \times 10^{-6}$

$= 12{,}566 \times 10^{-6}\ m^2$

$\rho = 2{,}82 \times 10^{-8}\ \Omega \cdot m$

$\epsilon = \rho \frac{I}{A} = (2{,}82 \times 10^{-8}) \frac{0{,}8}{12{,}566 \times 10^{-6}}$

$= (2{,}82 \times 10^{-8}) \times (63{,}66 \times 10^3) \approx 1{,}8\ V/m$

4). consider a hollow cylinder of length L and inner radius a and outer radus b. The material has resistivity ρ.

a). suppose a potential difference is applied between the ends of the cylinder and produces a current flowing parallel to the axis ?

b). IF instead the potential difference is applied between the inner and outer surfaces so that current flows radially outward, what is the resistance measured ?

Jawaban

a). $A = \pi (b^2 - a^2)$

$R = \rho \frac{L}{A}$

$R = \rho \frac{L}{\pi (b^2 - a^2)}$

$R_{total} = \frac{\rho L}{\pi (b^2 - a^2)}$

b). $dA = 2\pi r L\, dr$, L adalah panjang silinder dan r adalah radius elemen diferensial

$dR = \frac{\rho dr}{dA} = \frac{\rho dr}{2\pi r L}$

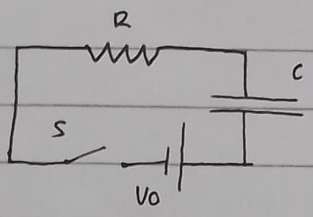
$R = \int_a^b \frac{\rho dr}{2\pi r l}$

$= \frac{\rho}{2\pi l} \int_a^b \frac{dr}{r}$

$$R = \frac{\rho}{2\pi l}\left(\ln(b) - \ln(a)\right)$$

$$= \frac{\rho}{2\pi l}\ln\left(\frac{b}{a}\right)$$

5).



Jika saklar ditutup pada saat $t = 0$, tentukanlah besar energy yang tersimpan dalam kapasitor (w_1) pada saat $t = \tau$ dan panas yang terdisipasi dalam resistor (w_2 sejak $t = 0$ hingga $t = \tau$

Jawaban

$$w_1 = \frac{1}{2} C V_c^2$$

- pada $t = \tau$, tegangan pada kapasitor V_c

$$V_c(\tau) = V\left(1 - e^{-\tau/(RC)}\right)$$

$$w_1 = \frac{1}{2}\left(CV^2(\tau)\right) = \frac{1}{2} C\left[V_0\left(1 - e^{-\frac{\tau}{RC}}\right)\right]^2$$

hukum joule $\longrightarrow$ $w_2 = \int_0^{\tau} p(t)\,dt$

$$p(t) = I(t)^2 R$$

$$I(t) = \frac{V_0}{R} e^{-\frac{t}{RC}}$$

$$p(t) = \left(\frac{V_0}{R} e^{-\frac{t}{RC}}\right)^2 R = \frac{V_0^2}{R} e^{-\frac{2t}{RC}}$$

$$w_2 = \int_0^{\tau} \frac{V_0^2}{R} e^{-\frac{2t}{RC}}\,dt$$

$$= \frac{V_0^2}{R} \cdot \frac{RC}{2}\left(1 - e^{-\frac{2\tau}{RC}}\right)$$

$$= \frac{V_0^2 C}{2}\left(1 - e^{-\frac{2\tau}{RC}}\right)$$

energi yg tersimpan dalam kapasitor pada waktu $t = \tau = w_1 = \frac{1}{2} C\left[V_0\left(1 - e^{-\frac{\tau}{RC}}\right)\right]^2$

- panas yang terdisipasi dalam resistor selama $t = 0$ hingga $t = \tau$ adalah $w_2 = \frac{V_0^2 C}{2}\left(1 - e^{-\frac{2\tau}{RC}}\right)$